Security Groups per RBAC Design

AccessGuard Health – Healthcare IAM Operations Simulation

Lab record: role-based security groups created in the HVMC.LAN domain

Project Name	AccessGuard Health
Project Type	Healthcare IAM Operations Simulation – Home Lab
Organization	Horizon Valley Medical Center (HVMC) – fictional
Board Task	Task 29 – Create Security Groups per RBAC Design (Completed)
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Verified On	November 2, 2026

1. Purpose

This document records the **security groups** created in the HVMC.LAN Active Directory domain from the RBAC (role-based access control) matrix designed in Week 1. In role-based access control, people are not given access one by one. They are placed in a group for their job, and the group holds the access. This is how the hospital applies **least privilege**: each person gets only what their role needs.

Everything here is a fictional training environment: no real patient data or real credentials are used.

2. Naming Convention

Every group follows one pattern so its purpose is clear from the name:

```
SG - <Department> - <Application> - <Access level>
example:  SG-Doctors-EHR-RW
```

Part	Meaning	Example
SG	Security Group	SG
Department	The role or team the group is for	Doctors
Application	The system the access is for	EHR (electronic health record)
Access level	RW = read and write; Read = read only; Admin = administrator	RW

3. Creating the Groups

The groups were created on the domain controller **hvmc-dc01** (192.168.56.10) with `samba-tool`, one command per group:

```
sudo samba-tool group add SG-Doctors-EHR-RW
sudo samba-tool group list
```

The first command creates a group. The second lists every group in the domain, including built-in ones, which is why the listing is filtered in the next section.

4. Verification: Full Group List

The list was checked on November 2, 2026 from the Mac Terminal over SSH. The filter shows only the groups that start with `SG-`:

```
ssh ubuntu@192.168.56.10
sudo samba-tool group list | grep "^SG-"
```

```
ubuntu@hvmc-dc01:~$ sudo samba-tool group list | grep "^SG-"
[sudo] password for ubuntu:
SG-Doctors-EHR-RW
SG-Reception-Scheduling-RW
SG-IT-ITAdmin-Admin
SG-Nursing-EHR-RW
SG-HR-HRPortal-RW
SG-Pharmacy-PharmacySystem-RW
SG-Lab-LIS-RW
SG-Contractors-VPN-Read
SG-Security-SecurityPortal-Admin
SG-Finance-Payroll-RW
ubuntu@hvmc-dc01:~$
```

Figure 1: The 10 role-based security groups in HVMC.LAN

Group	Department	Application	Access
SG-Doctors-EHR-RW	Doctors	EHR	Read/write
SG-Nursing-EHR-RW	Nursing	EHR	Read/write
SG-Reception-Scheduling-RW	Reception	Scheduling	Read/write
SG-IT-ITAdmin-Admin	IT	IT administration	Admin
SG-HR-HRPortal-RW	HR	HR portal	Read/write
SG-Pharmacy-PharmacySystem-RW	Pharmacy	Pharmacy system	Read/write
SG-Lab-LIS-RW	Lab	Laboratory information system (LIS)	Read/write
SG-Contractors-VPN-Read	Contractors	VPN	Read only
SG-Security-SecurityPortal-Admin	Security	Security portal	Admin
SG-Finance-Payroll-RW	Finance	Payroll	Read/write

5. Security Considerations

- All names, departments and users are **fictional**; no real patient or employee data is used.
- Access is given through **groups, not individuals**, so changing a person's role means changing group membership (the "mover" process).
- Contractors receive **read-only** VPN access; only IT and Security hold **admin** groups (least privilege).

- Group names make audits easier: anyone reading a group name can see who it is for and how much access it gives.

6. Key Outcomes

- Created 10 role-based security groups following one naming convention.
- Verified the full group list from the server.

7. Next Steps

- Task 38: run the onboarding activity (create a new user and assign the right group).
- Attach a real resource to a group (for example, a shared folder only SG-Doctors-EHR-RW can open) to show the access in action.

8. Deliverables

- Security Groups per RBAC Design: Lab Documentation for GitHub Portfolio (this document).
- Screenshot of the full group list (Figure 1).